

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
0	1	55	Male	Blouse	Clothing	53
1	2	19	Male	Sweater	Clothing	64
2	3	50	Male	Jeans	Clothing	73
3	4	21	Male	Sandals	Footwear	90
4	5	45	Male	Blouse	Clothing	49

	Location	Size	Color	Season	Review Rating	Subscription Status
0	Kentucky	L	Gray	Winter	3.1	Yes
1	Maine	L	Maroon	Winter	3.1	Yes
2	Massachusetts	S	Maroon	Spring	3.1	Yes
3	Rhode Island	M	Maroon	Spring	3.5	Yes
4	Oregon	M	Turquoise	Spring	2.7	Yes

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	Express	Yes	Yes	14
1	Express	Yes	Yes	2
2	Free Shipping	Yes	Yes	23
3	Next Day Air	Yes	Yes	49
4	Free Shipping	Yes	Yes	31

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly
2	Credit Card	Weekly

3	PayPal	Weekly
4	PayPal	Annually

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3900 entries, 0 to 3899
```

```
Data columns (total 18 columns):
```

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	object
17	Frequency of Purchases	3900 non-null	object

```
dtypes: float64(1), int64(4), object(13)
```

```
memory usage: 548.6+ KB
```

```
#Summary statistics using .describe()
```

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	\
count	3900.000000	3900.000000	3900	3900	3900	
unique	NaN	NaN	2	25	4	
top	NaN	NaN	Male	Blouse	Clothing	
freq	NaN	NaN	2652	171	1737	
mean	1950.500000	44.068462	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	

	Purchase Amount (USD)	Location	Size	Color	Season	Review	\
Rating							
count	3900.000000	3900	3900	3900	3900		

3863.000000					
unique	NaN	50	4	25	4
NaN					
top	NaN	Montana	M	Olive	Spring
NaN					
freq	NaN	96	1755	177	999
NaN					
mean	59.764359	NaN	NaN	NaN	NaN
3.750065					
std	23.685392	NaN	NaN	NaN	NaN
0.716983					
min	20.000000	NaN	NaN	NaN	NaN
2.500000					
25%	39.000000	NaN	NaN	NaN	NaN
3.100000					
50%	60.000000	NaN	NaN	NaN	NaN
3.800000					
75%	81.000000	NaN	NaN	NaN	NaN
4.400000					
max	100.000000	NaN	NaN	NaN	NaN
5.000000					

	Subscription Status	Shipping Type	Discount Applied	Promo Code
Used \				
count	3900	3900		3900
3900				
unique	2	6		2
2				
top	No	Free Shipping		No
No				
freq	2847	675		2223
2223				
mean	NaN	NaN		NaN
NaN				
std	NaN	NaN		NaN
NaN				
min	NaN	NaN		NaN
NaN				
25%	NaN	NaN		NaN
NaN				
50%	NaN	NaN		NaN
NaN				
75%	NaN	NaN		NaN
NaN				
max	NaN	NaN		NaN
NaN				

	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900	3900

unique	NaN	6	7
top	NaN	PayPal	Every 3 Months
freq	NaN	677	584
mean	25.351538	NaN	NaN
std	14.447125	NaN	NaN
min	1.000000	NaN	NaN
25%	13.000000	NaN	NaN
50%	25.000000	NaN	NaN
75%	38.000000	NaN	NaN
max	50.000000	NaN	NaN

#Checking if missing data or null values are present in the dataset
`df.isnull().sum()`

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

`dtype: int64`

#Renaming columns according to snake casing for better readability and documentation

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

`df.columns`

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
# create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

```
# create new column purchase_frequency_days
```

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
```

```
df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes

1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
# Dropping promo code used column
```

```
df = df.drop('promo_code_used', axis=1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group',
      'purchase_frequency_days'],
      dtype='object')
```

```
pip install psycopg2-binary sqlalchemy
```

```
Defaulting to user installation because normal site-packages is not
writeable
```

```
Collecting psycopg2-binary
```

```
  Downloading psycopg2_binary-2.9.11-cp313-cp313-
win_amd64.whl.metadata (5.1 kB)
```

```
Requirement already satisfied: sqlalchemy in c:\programdata\anaconda3\
lib\site-packages (2.0.39)
```

```
Requirement already satisfied: greenlet!=0.4.17 in c:\programdata\
anaconda3\lib\site-packages (from sqlalchemy) (3.1.1)
```

```
Requirement already satisfied: typing-extensions>=4.6.0 in c:\
programdata\anaconda3\lib\site-packages (from sqlalchemy) (4.12.2)
```

```
Downloading psycopg2_binary-2.9.11-cp313-cp313-win_amd64.whl (2.7 MB)
```

```
----- 0.0/2.7 MB ? eta -:--:--
```

```
----- 0.0/2.7 MB ? eta -:--:--
```

```
----- 0.5/2.7 MB 6.6 MB/s eta
```

```
0:00:01
```

```
----- 2.7/2.7 MB 9.1 MB/s eta
```

```
0:00:00
```

```
Installing collected packages: psycopg2-binary
```

```
Successfully installed psycopg2-binary-2.9.11
```

```
Note: you may need to restart the kernel to use updated packages.
```

```
from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
# Replace placeholders with your actual details
username = "postgres"      # default user
password = "root" # the password you set during installation
host = "localhost"        # if running locally
port = "5432"             # default PostgreSQL port
database = "customer_behavior" # the database you created in
pgAdmin

engine = create_engine(f"postgresql+psycopg2://{username}:
{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer1"  # choose any table name

print(f"Data successfully loaded into table '{table_name}' in database
'{database}'.")

Data successfully loaded into table 'customer1' in database
'customer_behavior'.
```